

SURV/SURVMETH 720

Multiple Sources of Survey Error

Class #11

November 24, 2025

Occupation coding

- Stand-out example of coding open-ended responses:
- Three methods:
 - Fully manual
 - Fully automatic
 - Computer-assisted

Using AI for coding open-ended responses

- But also to: probe when an insufficient open-ended response received
 - Sturgis, 2025 (AAPOR 2025): <https://tinyurl.com/LSEps25> (~80% agreement with manual coding)
 - Geisen et al. (2025). Respondents seem to find follow-ups more engaging.

Some general directions

- ‘Few shot’ prompting > ‘Zero shot’ prompting
- Prompts important. Ideally provide both positive and negative examples
- Pretraining has been found to be necessary
- Performance differs between LLM/NLP models
- Gweon and Schonlau, 2024:
 - BERT for coding open-ended responses
 - Find that fine-tuning the model is essential
 - If manually trained on 100 observations, BERT barely beats non-pretrained machine learning approaches
 - Advantage rapidly increases with more coded observations (e.g., 200-400)

What do you think and likely challenges and pitfalls in using AI for coding open-ended responses?

This week's focus: Multiple
sources of error

MSE in the TSE context

- Aggregating error
 - Provides information to data users regarding the overall quality of reported estimates
 - Example: $MSE(\bar{y}) = B_{NR}^2 + B_{Meas}^2 + Var_{Sam}$
 - Incomplete, but potentially useful
- Decomposing error
 - Useful for understanding and targeting major sources of error
 - Example: decomposing *selection* error into coverage error and nonresponse error
 - $B(\hat{\theta}) = B_{Cov} + B_{NR}$

Studies of multiple sources of survey error:

Key objectives

- Need to understand the relative magnitudes of each error source in various survey estimates; optimize resource allocation to improve estimates
- Error to think that error sources are always independent.
- Need to understand the interplay between errors as a function of survey design features; e.g., are additional call attempts beneficial ?
- Ideally, obtain a causal understanding of the relationship between survey errors

Challenges

- Typically, no repetition
- Study of a source of error often requires a special design.
 - What do you need to estimate non-response bias?
 - What do you need to estimate measurement error?
 - What do you need to gauge the impact of two different questionnaire wordings?
- The possible association between sources of error requires designs that allow **simultaneous** estimation of the multiple sources of error under investigation.

1. How do we get estimates of bias and variance in the absence of replication?

Shirani-Mehr et al (2018)

- “...quantify the uncertainty in predictions left unmeasured by traditional margins of errors.”
- Used 4,221 polls for 608 state-level presidential, senatorial, and gubernatorial elections between 1998 and 2014
- Decomposed error into election-level bias and variance terms.
- Estimated average election-level error
 - RMSE $\sim$ 3.5 % points, approx. 2 x reported MoE

2. Is unit non-response related to item non-response ?

Fricker and Tourangreau (2010)

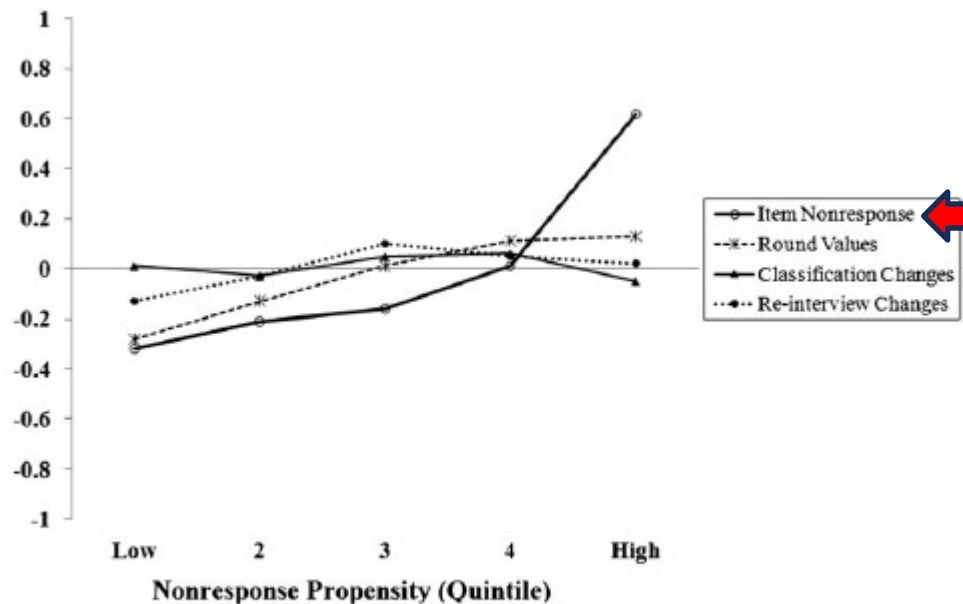
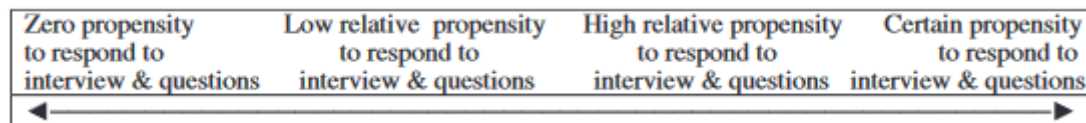


Figure 1. Relationship of CPS Data-quality Indicators (in Standard Deviation Units) to CPS Nonresponse Propensity.

- Add to the model:
 - hours worked, commute time (proxy for busyness)
 - Item burden (proxies for respondent effort)
 - Marital status, Home ownership, etc.. (proxies for social capital)
- But the relationship between unit and item NR remains.
- Why? What could be a common cause?

- Motivation model: “Individuals who lack motivation are simultaneously less likely to participate in a survey and, when eventually convinced to participate, have lower commitment to the respondent task “
 - Applies more specifically to data quality issues that have a motivational component (item non-response) especially on burdensome or difficult questions.
- Converted refusers are prone to higher levels of item nonresponse than other respondents both for specific items and overall (Olson 2013)
- Response continuum model (Yan and Curtin 2010)



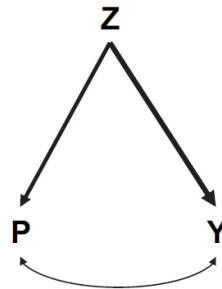
3. More generally, are reluctant respondents also poor respondents ?

Conceptual models of correlation between NRE and ME

- Independence model: NR due to situational or motivational factors but ME largely due to cognitive factors. Thus, errors are thought to be generated independently.
- But is that the case ? Remember the satisficing formula. Hence the 'common cause' model.



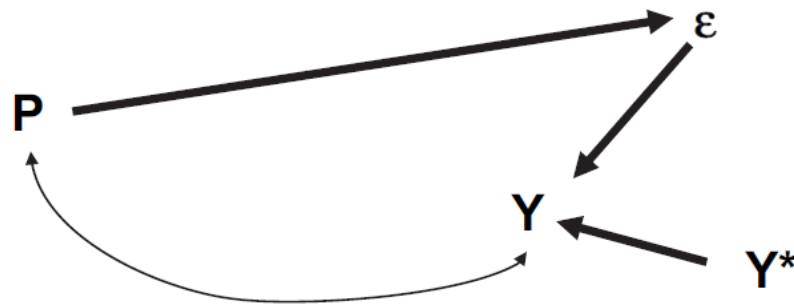
1. Separate Causes Model



2. Common Cause Model



3. Survey Variable Cause Model



4. Nonresponse-Measurement Error Model

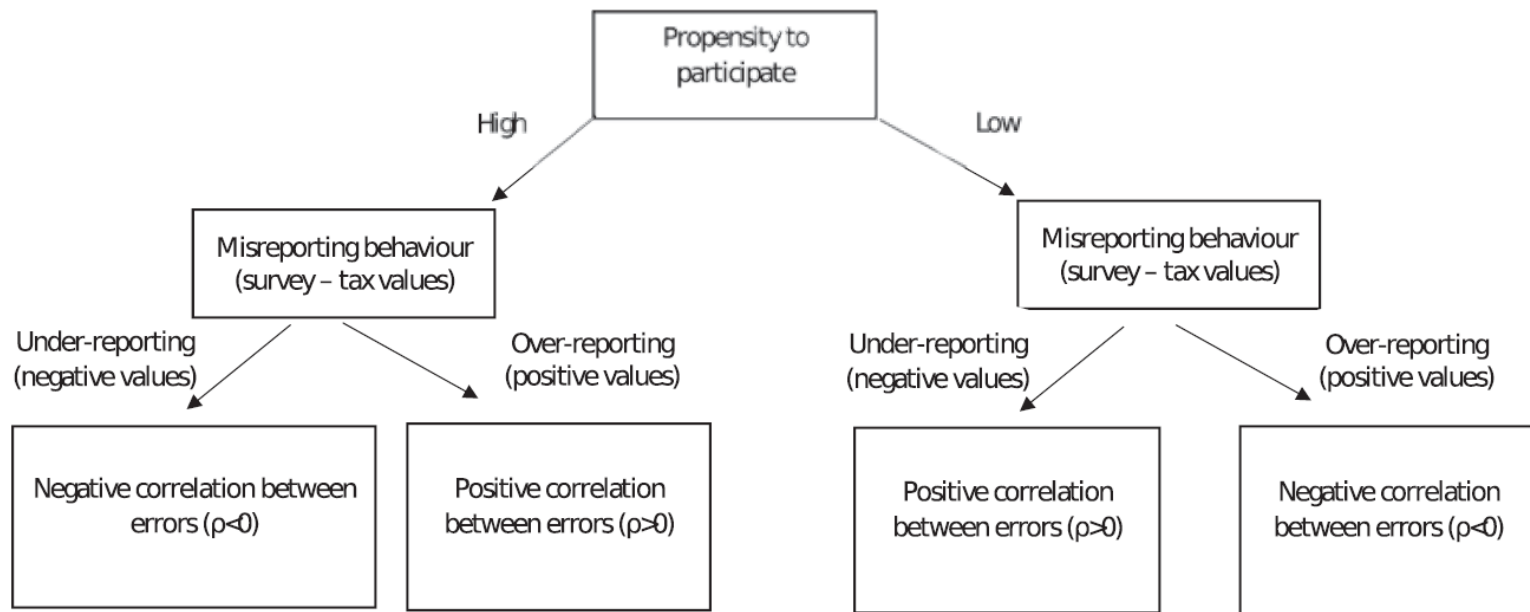


Figure 1. Correlation between Participation Propensity and Income Misreporting.

Source: Neri and Porreca, 2024 (JSSAM)

Several hypothesis re. the link between level of effort (#contact attempts, refusal conversion, etc.) and ME (*see Olson 2013*)

- Motivation (covered earlier)
- Reactance
- Interest in the survey topic
- Interest in the sponsor
- Personal characteristic/Composition
- Research importance
- Self-perception

Cannell & Fowler (1963)

- Measurement error increases for respondents who are more difficult to recruit (motivation?)

TABLE 6
REPORTING OF HOSPITAL EPISODES AND ACCURACY OF REPORTING LENGTH
OF STAY, BY WAY IN WHICH FOLLOW-UP REPORT WAS OBTAINED
IN SELF-ENUMERATIVE PROCEDURE

	<i>Way in Which Follow-up Was Obtained</i>			<i>Total</i>
	<i>1st Mail Form</i>	<i>2nd Mail Form</i>	<i>Telephone or Personal Visit</i>	
Number of hospital records	394	75	77	546
Number of episodes reported by respondent	343	64	52	459
Per cent of sample episodes not reported	13	15	32	16
Per cent of reports in which length of stay was correct	60	53	33	57

Assumption in the 3rd data column?

Peytchev, Peytcheva, & Groves (2010)

- Do reluctant respondents disproportionately fail to report abortions ?
- NSFG 2002-2003
- Two-phase field work: 2nd phase focused on non-response (reducing non-response + improving sample balance)
- Measure abortion in CAPI and ACASI mode – latter used as comparative standard.

Peytchev at al ...

Estimate	Response Propensity Quintiles				
	1 Lowest Propensity	2	3	4	5 Highest Propensity
Among all respondents, $n = 7,547$					
Percentage reporting at least one abortion in ACASI, but none in CAPI	3.8% (0.7)	2.2% (0.4)	2.2% (0.5)	2.3% (0.5)	2.0% (0.4)
Among first-phase reporters of at least one abortion in ACASI, $n = 1,287$					
Percentage reporting at least one abortion in ACASI, but none in CAPI	23.7% (4.1)	13.2% (2.4)	10.3% (2.0)	13.3% (2.5)	11.8% (2.1)

Standard errors in parentheses

Effects remain even after adjusting for covariates like age, race, and education

- So, should they **not** have spent efforts recruiting the bottom quintile samples ? Seems so – higher RMSE.

- What about the 2nd phase respondents?

Low sample size but some evidence that link between NRE and ME can be broken.

Applicable theory (common cause): Self-perception (social desirability)

Compare with Neri and Porreca (2024; JSSAM):

- Italian Survey on Income and Wealth linked with administrative income data from tax returns
- Find a positive correlation between the unit NR and ME
 - Households at the extremes of the income distribution mainly cause this association.
- $ME > NR$
- Efforts to reduce nonresponse rates are worthwhile only for nonrespondents in the lowest estimated response propensity group.
 - If these households participate, the bias decreases because of the reduction in NR that offsets the increase in ME.
 - Practical implication: Need a targeted approach to reducing non-response

Neri and Porreca, 2024...

Likely common causes

- High earners: Low propensity, concerns with confidentiality
- Low earners: High propensity, social desirability

But mixed results...

- Hox, de Leeuw, and Chang (2012; Bulletin of Sociological Methodology): validity and reliability of well-being questions do not vary between eager and reluctant respondents.
- Olson 2006; Reverse record check: Wisconsin Divorce Study Divorce certificates. One member of the divorced couple selected at random to be the R.
- CATI RR = 71%
- Analysis:
 - Nonresponse bias estimate: respondent means vs entire frame means (using records)
 - Measurement error bias estimate : Respondent mean using records vs and respondent mean using self-reports
 - Propensity quintiles : NRE vs ME in each quintile

Olson 2006...

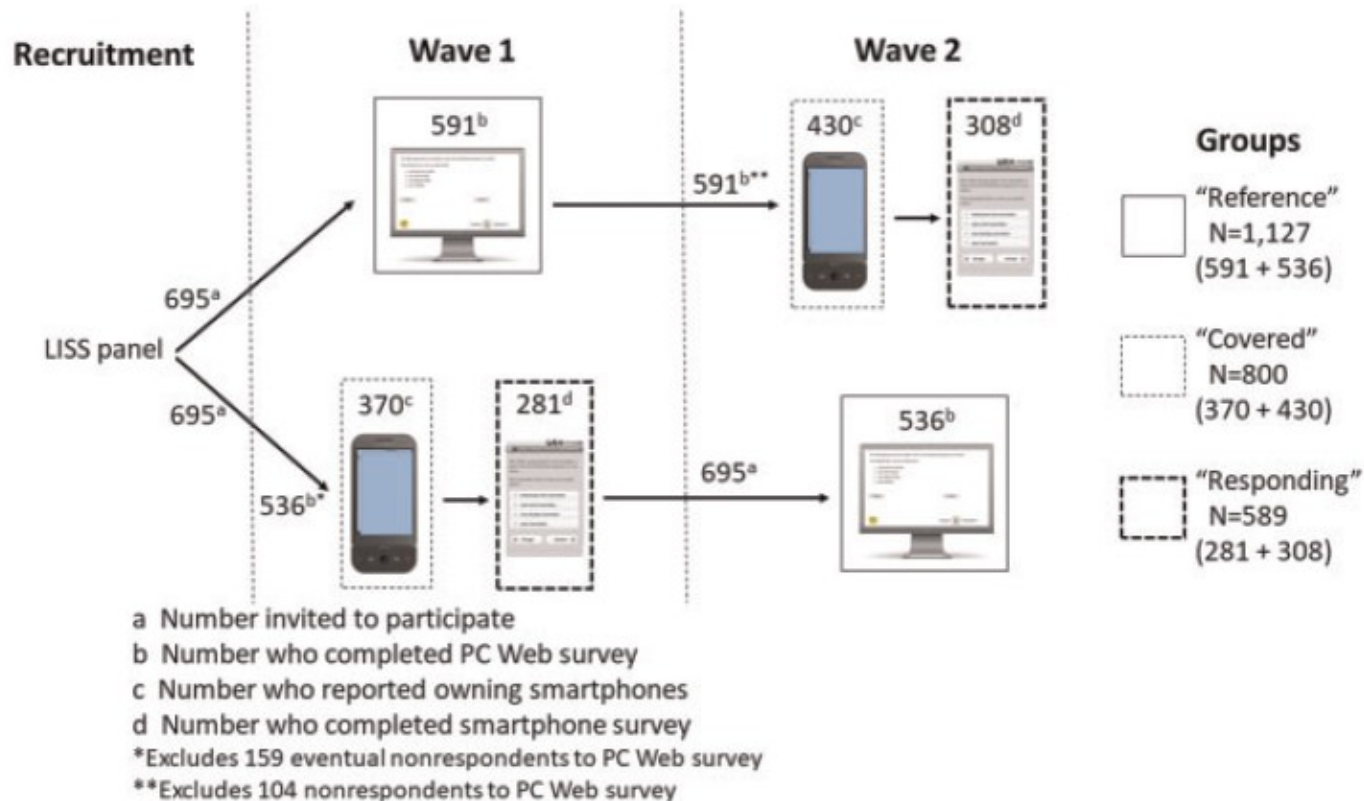
- Limited support was found for the suspicion that measurement error bias increases for estimates including reluctant respondents.
- Nonresponse bias larger than measurement bias for 2 of 3 estimates.

	Contact					Cooperation				
	Magnitude of Bias			% Contribution to Total Bias		Magnitude of Bias			% Contribution to Total Bias	
	ME	NR	Total	ME	NR	ME	NR	Total	ME	NR
	1	2	3	4	5	6	7	8	9	10
Length of Marriage										
High	2.50	1.86	4.36	57%	43%	2.09	14.73	16.82	12%	88%
4	2.11	7.52	9.63	22%	78%	2.49	12.43	14.92	17%	83%
3	0.35	8.91	9.26	4%	96%	0.41	9.03	9.44	4%	96%
2	0.21	6.53	6.74	3%	97%	1.52	4.68	6.20	25%	75%
Low	0.24	3.88	4.12	6%	94%	0.24	3.88	4.12	6%	94%
Time Since Divorce										
High	6.72	0.34	7.06	95%	5%	3.86	0.03	3.89	99%	1%
4	6.11	1.76	7.87	78%	22%	5.05	0.12	5.16	98%	2%
3	6.30	1.08	7.38	85%	15%	5.47	1.47	6.94	79%	21%
2	5.71	1.23	6.94	82%	18%	5.23	0.83	6.07	86%	14%
Low	5.30	0.69	5.99	89%	11%	5.30	0.69	5.99	89%	11%
Number of Marriages										
High	0.007	0.053	0.061	12%	88%	0.009	0.029	0.037	23%	77%
4	0.021	0.054	0.075	28%	72%	0.000	0.046	0.046	0%	100%
3	0.010	0.027	0.037	27%	73%	0.008	0.033	0.042	20%	80%
2	0.011	0.029	0.040	28%	72%	0.013	0.021	0.034	39%	61%
Low	0.011	0.025	0.036	30%	70%	0.011	0.025	0.036	30%	70%

4. Is what seems as a measurement issue actually due to other error sources ?

Antoun et al. (2019)

Cross-over design



- 2-wave survey using PC Web responses used as a benchmark

- Smartphone-survey produced biased estimates relative to the PC web reference survey for a subset of the survey variables. The average size of the mode effects —2.3 percentage points—was not large.
- Decomposition shows that coverage error to be the largest contributor of error in spite of relatively high coverage rates (71 percent).
- No evidence of measurement effects, at least for the types of questions they examined; conditional on coverage and response to the survey, respondents gave similar responses on smartphones as they did on PCs.

5. More on modes + item characteristics

Sakshaug, Yan, & Tourangeau (2010)

- Somewhat reverse fashion -> do efforts at reducing measurement error increase non-response error ?
- Survey of UMD alumni (admin records available) using CATI, IVR, and a web survey.
- **Socially undesirable items:** Measurement bias (8 - 20 percentage points) > NR bias (2 - 3 points)
- **Socially desirable and neutral items:** NR bias (0 - 15 points) > measurement bias (0 - 7 points)
- Switching to self-administered modes can reduce ME but increase overall error due to drop-offs.
- Additional call-backs reduce noncontact bias but less consistent relationship with ME.

Table 1. Percentage/Mean in Each Subgroup, According to Frame and Survey Data (standard errors in parentheses)

	Before Mode Switch Frame Data			After Mode Switch Frame Data		Survey Report
	Sample (<i>n</i> = 7,535)	Contacts (<i>n</i> = 3,497)	Screeners <i>Rs</i> (<i>n</i> = 1,501)	Mode Switch <i>Rs</i> (<i>n</i> = 1,107)	Item Responders (<i>n</i> 's vary)	Survey Data Item Responders (<i>n</i> 's vary)
Undesirable Characteristics						
GPA < 2.5	15.3 (0.4)	14.9 (0.6)	13.7 (0.9)	12.2 (1.0)	12.1 (969; 1.1)	4.2 (969; 0.7)
At least one D/F	62.6 (0.6)	62.4 (0.8)	61.0 (1.3)	61.1 (1.5)	60.7 (1,071; 1.5)	45.5 (1,071; 1.5)
Dropped a class	70.9 (0.5)	69.7 (0.8)	69.3 (1.2)	68.7 (1.4)	68.1 (1,057; 1.4)	47.9 (1,057; 1.5)
Desirable Characteristics						
GPA > 3.5	18.6 (0.5)	19.5 (0.7)	20.9 (1.1)	21.4 (1.2)	22.0 (969; 1.3)	23.1 (969; 1.3)
Honors	9.4 (0.3)	9.9 (0.5)	11.3 (0.8)	11.9 (1.0)	12.3 (1,062; 1.0)	17.2 (1,062; 1.2)
Ever donated to UMD	25.3 (0.5)	30.9 (0.8)	38.0 (1.3)	40.0 (1.5)	40.6 (1,019; 1.5)	41.4 (1,019; 1.5)
Donated in last year	8.5 (0.3)	11.6 (0.5)	15.1 (0.9)	15.4 (1.1)	15.7 (1,001; 1.2)	17.1 (1,001; 1.2)
Alumni member	7.1 (0.3)	9.8 (0.5)	14.5 (0.9)	15.9 (1.1)	15.9 (1,048; 1.1)	23.2 (1,048; 1.3)

Sakshaug, Yan, & Tourangeau (2010)
Only part of Table 1 shown here

Tourangeau, Groves, Redline (2010)

- Hypothesis:
 - those in a socially undesirable category (nonvoters) would be less likely to take part in a survey that might include questions about that topic +
 - they would also be more likely to misreport about it if they did take part in the survey
- Testing common cause (past voting; social desirability) and survey topic theories + intervention (re:Peytchev et al 2010).
- Fully crossed 3-factor experiment: survey topic description (politics vs health), mode (mail vs telephone), and incentive
- Sample voters and non-voters from a list of registered voters

Tourangeau, Groves, Redline (2010)

Table 5. Estimated Percentage of Voters in the 2004 and 2006 Elections (sample sizes in parentheses) by Experimental Variables and Voter Status

	Entire sample (Frame data)	Respondents (Frame data)	Respondents (Survey report)	Bias estimates	
				Nonresponse	Measurement
Voted in 2004					
Overall	47.6 (2689)	58.3 (904)	80.1 (890)	10.7	21.8
Politics, elections, & voting	47.1 (1346)	60.3 (441)	81.3 (438)	13.2	21.0
Health & lifestyles	48.2 (1343)	56.4 (463)	79.0 (452)	8.2	22.6
\$5 incentive	47.2 (1349)	55.8 (591)	80.3 (584)	8.6	24.5
No incentive	48.1 (1340)	62.9 (313)	79.7 (306)	14.8	16.8
Telephone	47.8 (1020)	60.3 (350)	83.0 (342)	12.5	22.7
Mail	47.5 (1669)	57.0 (554)	78.3 (548)	9.5	21.3

Tradeoff/Interplay between Error Sources

- Design features may affect more than one source of error.
- TSE perspective: base design decisions on overall effects - not just their effects on one source of error.
- Since bias is directional, individual errors can:
 - accentuate each other, such that net bias $>$ sum of absolute biases
 - be independent
 - cancel each other out, such that net bias $\leq$ sum of absolute biases
- Illustration: <https://bit.ly/3gg0N10>

Size of Different Errors

- Which error is the largest contributor to total error?
 - Difficult to say in any general sense
 - May vary across surveys
 - May vary across estimates within the same survey

- Different methods: Frame data (Olson 2006), relative bias (Peytchev), level of effort (Cannell and Fowler)
- What happens to statistics other than means ?
- What do these tell us about earlier studies that focused only on single error sources?

Next Week

- TSE and found data
- Questions?